**Water Monitoring Buoy**
Sponsor: Self, No sponsor (Senior Design Student)

**Background:**
Communities near rivers, ponds and agricultural watersheds may lack real time coverage of water conditions and flood risk. Water quality degradation caused by agricultural runoff and other forms of contamination are only recognized after they occur. This results in a lack of maximum production, public health risks and flash floods, being detected after containment efforts can even be made. Continuous automated environmental monitoring systems capable of detecting these issues already exist but can be unaffordable for small communities that cannot afford current monitoring systems. This project aims to develop a low cost, autonomous alternative that can provide early warning detection for small communities, agricultural practices and ecosystem management.

**Project Needs:**
1. Multi Parameter Environmental Dataset: Continual collection of calibrated water quality data (temperature, pH, turbidity, conductivity and dissolved oxygen) and weather relevant data (water depth, rainfall, barometric pressure) from a site where the system has been deployed as baseline.
2. Flood risk and Anomaly Modeling: Applying statistical analysis and machine learning to the collected data to detect anomalies that may have occurred to provide warning to future indications. Allowing for the system to improve as more data is collected.

**Project Scope:**
1. Sensor Integration and Calibration: Chosen sensors must integrate together with the system at it's sampling speed for proper data collection.
2. Embedded Firmware: Utilize high speed microcontroller to sample data, conduct power management and for wireless communication.
3. Wireless Data Pipeline: Extract data wirelessly and forward data to the cloud.
4. Cloud Backend and Dashboard: Develop simplified dashboard providing data and diagnostics, allow for SMS and email sending of alerts.
5. Machine Learning: Implement anomaly detection.
6. Field Deployment: Design system to be waterproof and deployable for various aquatic environments.

**Potential Technical Considerations:**
1. Power Budget: Energy consumed by the system and energy stored.
2. Sensor Calibration: As the sensor operates it may drift and need recalibration.
3. Wireless Range and Reliability: The wireless communication must be able to interact reliably within various distances.
4. Limited Data For Training: Some data publicly available could be used but will heavily focus on data collected during the construction phase.

5. Environmental Durability: Must protect electronics from water intrusion, corrosion and temperature change throughout the seasons.

**Deliverables:**
1. Functional Prototype: Capable of recording necessary data and system diagnostics.
2. Cloud Platform: Data will be stored and viewable from the cloud.
3. Technical Documentation: Comprehensive report including functionality and operation.
4. Project Report and Presentation: The finalized report summarizing findings along with a presentation demonstrating the system's capabilities.

**Assumptions and Dependencies:**
1. Access to permitted deployment sites for field testing.
2. Availability of project budget for sensors, communication, hardware and enclosure.
3. Availability of cloud hosting resources.

**Why Choose This Project?:**
1. Real World Impact: Contribution to affordable agricultural, environmental data collection and community early warning flood detection.
2. Full stack development incorporating electrical and computer and software engineering from analog sensors, embedded firmware and cloud/ML applications.